

RENANCE

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

EMT 411 — Past Questions & Materials
by Renance

Course: EMT 411
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 2
Levels: 400

This study pack was compiled and restructured by Renance for EMT 411 students at Federal University of Petroleum Resources, Effurun. It brings together 2 past paper(s)/material(s) into one clean, readable document so you can revise without chasing scattered PDFs. Practise these same questions interactively on Renance with instant answers, AI explanations, and full CBT mock exams — over 235,000+ questions across JAMB, Post-UTME, WAEC, NECO and university courses.

Study online: renance.app
Get the app: Google Play — [dev.renance.renance](https://play.google.com/store/apps/details?id=dev.renance.renance)

Generated by Renance AI

Paper 1 of 2 — EMT 411 environmental monitoring system and techniques. 2024/2025

Level 400 • First Semester • PastQuestion

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

COLLEGE OF SCIENCE

ENVIRONMENTAL MANAGEMENT AND TOXICOLOGY

FOR 2024/2025 ACADEMIC SESSION

Monitoring Systems and Techniques (3Units)

DEPARTMENT OF ENVIRONMENTAL ENGINEERING

FIRST SEMESTER EXAMINATION

COURSE TITLE: Environmental monitoring

~ COURSE CODE: EMT 411

INSTRUCTIONS: ATTEMPT FOUR QUESTIONS

(SECTION)

TIME: 2 HOURS

ANSWERS (AT LEAST ONE FROM EACH

SECTION A

1a) Differentiate between point source and non-point source pollutant?

b) (i) What are criteria pollutants?

Gi) List six (6) USEPA designated criteria pollutants

technologies in large-scale environmental

remote sensing satellite-based data collection?

a)

pollutants.

c) Assess the importance of remote sensing technology

monitoring. What are the advantages and limitations of

remote sensing in air quality and atmospheric

pollution? ==) Explain the application of LIDAR technology

in air quality monitoring.

2. a) Briefly state six (6) steps that you need to follow when carrying out a successful air

quality assessment. 7

person Billi

quality assessment. 7

5) i) What is Ambient Air-monitoring.

ii) Briefly categorize Ambient Air-monitoring.

3. a) Evaluate the death of plant foliage (5 km

radius) you are consulted to evaluate

fumes. Using a

c) As an environmentalist,)

radius) around a factory producing SO₂ (Sulphur Dioxide) as waste emissions from the factory that led

balance equation, discuss the phenomenon caused by

to the death of the plants. fob! 4

d) Differentiate between passive and active advantages and limitations in air quality monitoring.

SECTION B gl?

air sampling techniques, highlighting their

3g. Define aquatic toxicology

3b. Discuss the different types of exposure techniques

3c. Write short notes on bioassays

Ja. What are the factors responsible for the susceptibility

Ab, List five common standard test species used in toxicity te

4c, What are the bases for their selection by ASTM?

5a. Define the following: Biomonitoring, LCs, EC₅₀, Bioaccumulation, Bioconcentration factor

Sb. The effects of environmental toxicants vary

toxicants involved. Explain

5f. Describe the different aquatic tests used in aquatic toxicology

of different species to chemicals?

sting.

depending on the different chemicals or

FEDERAL

UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

COLLEGE OF SCIENCE

DEPARTMENT OF ENVIRONMENTAL

FIRST SE

COURSE TITLE: Environmental monitoring

—~COURSE CODE: EMT 411

INSTRUCTIONS: ATTEMPT FOUR QUESTIONS (AT LEAST ONE

SECTION)

MANAGEMENT AND TOXICOLOGY

MASTER EXAMINATION FOR 2024/2025 ACADEMIC SESSION

Monitoring Systems and Techniques (3Units)

TIME: 2 HOURS

NE FROM EACH

SECTION A

1a) Differentiate between point source and

b) (i) What are criteria pollutants?

(ii) List six (6) USEPA designated criteria pollutants.

g technologies in

limitations of satellite-based data

c) Assess the importance of remote sensing

monitoring. What are the advantages and

disadvantages of LIDAR technology in air quality monitoring.

2.a) Briefly state six (6) steps that you need for a quality assessment.

b) i) What is Ambient Air-monitoring.

ii) Briefly categorize Ambient Air-monitoring.

Ited to evaluate the data

1g SO₂ (Sulphur Dioxide) as waste fumes. Using a

c) As an environmentalist, you are considering

radius) around a factory producing

balance equation, discuss

to the death of the plants.

d) Differentiate between

advantages and limitations

discuss the phenomenon of ozone

passive and active air

as in air quality monitoring.

non-point source pollutant?

large-scale environmental

a collection?

technology in air quality and atmospheric

steps to follow when carrying out a successful air

death of plant foliage (5 km

led by emissions from the factory that led

sampling techniques, highlighting their

SECTION B

3a. Define aquatic toxicology

3b, Discuss the different types of exp
3c, Write short notes on bioassays
4a. What are the factors responsible for the s
Ab, List five common standard test species u
Ac, What are the bases for their selection by
5a. Define the following: Biomonitoring,
factor
Sb. The effects of environmental toxicants Vv
toxicants involved, Explain
5c, Describe the different aquatic tests used
osure techniques
usceptibility of different species to chemicals?
sed in toxicity testing.
ASTM?
LCos, ECio, Bioaccumulation, Bioconcentration
ary depeding on the different chemicals or
in aquatic toxicology

Paper 2 of 2 — EMT 411 ENVIRONMENTAL MONITORING SYSTEMS 2019,2020

Level 400 • First Semester • PastQuestion

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN »

COLLEGE OF SCIENCE

DEPARTMENT OF ENVIRONMENTAL MANAGEMENT AND TOXICOLOGY

FIRST SEMESTER EXAMINATION FOR 2019/2020 ACADEMIC SESSION

COURSE TITLE: ENVIRONMENTAL MONITORING SYSTEMS —2 Units

COURSE CODE: EMT 411 TIME: Two Hours

INSTRUCTION: Answer four questions, two from each Section

SECTION A

ie (a) State the objectives of environmental monitoring

(b) Briefly explain one of the three major methods of monitoring the environment.

2, Write short note on the following

5 3 e an

(a) Bioconcentration ~ Vest

é prope “poe C=

(b) Biomagnification —

(c) Bioaccumulation

(d) Bioindicators

35 The substances released into the environment as a consequence of human activities can be classified in two main categories. Mention and briefly explain them.

SECTION B

4a. Briefly justify your understanding of the term Air Pollution Meteorology with 2 case studies.

4b. Briefly explain “Ozone” stating the process of formation. "y id Can, i

ow Fane nus

ap

5a. Mention 3 biological air pollutants and enumerate 3 ways of mitigating exposure to

“A>

them OEE gn del

AMMoo ver >

5b. Identify 5 symptoms of health problems caused by biological pollutants

6a. _ Briefly explain the mechanism of operation of the following air pollution control

y ~ equipment. concer faye er 5s bie

a) Scrubbers b) Cyclones c) Air filters d) Biofilters Akh ys

an sxs AUkeC Premera

=) Laval by Ou u F

gel ePA cal s > 3 Btdan | sng hoy ¢ Meu \

Ae + Bayete. ue = \$e & afta

we

: ee Ne S Lichen,

(o\c\ins wh "e

Cal a LQ kaya)

4 (Cal Ga